A Review on Supervised Machine Learning Text Categorization Approaches

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***Abstract*— The inception of abundant and hefty digitized text documents has emerged in recent years. Such texts need to be categorized technically for storage and retrieval purpose. Text Categorization Machine Learning techniques exist to manage the substantial data. These techniques can be divided into Supervised and Unsupervised Machine Learning techniques. In this paper we review Supervised Machine learning techniques that deal with generating patterns and hypothesis according to the prior knowledge. The objective of this paper is to highlight important features and methods of Supervised Machine Learning techniques. Selecting important features and then classifying the text is necessary for effective text categorization. The performance of Text Categorization and dimensionality improves by performing feature selection in the existing techniques. The importance of feature selection is considered in this paper. The general comparison table of techniques is presented for in depth parametric assessment.**

Keywords*— *Supervised, Machine Learning, Text Categorization, Text Classification, K Nearest Neighbour, Support Vector Machine, Feature Selection, Decision tree, Random Forest, Rocchio, Bayesian.

I. INTRODUCTION

In today's era, due to increase in digitized text documents, categorization or classification has become vital for certain purpose like storage, filtering etc. Text Categorization means assigning categories to unlabeled documents [8]. Each document can have single, multiple or no categories [11]. Categorizing text has become a major problem in many domain. There are many applications of Text Categorization that makes it important to be analyzed and studied. Applications like spam filtering especially in emails, automatic grading of essays, automatic indexing of scientific articles or documents, classifying news articles and magazines, survey coding to name a few [12]. Moreover, Text Categorization plays dominant role in sentiment analysis, information retrieval, text summarization and medical diagnosis [13]. Due to large amount of data, complexity arises when we try to classify the data. Some techniques have been derived to lower the complexity of categorizing text. One of the Intelligence techniques called Machine Learning techniques plays eminent role in classifying text or data. Machine Learning techniques construct a classifier by learning from the set of unclassified documents based on the characteristics of categories [10]. Such Machine Learning techniques are of two types namely Supervised Machine Learning Technique and Unsupervised Machine Learning Technique. Supervised Machine Learning techniques are those in which the sample data or instances are supplied externally. Using those instances, it produces general patterns and hypothesis. In general, Supervised Machine Learning algorithm works on prior information. Moreover, Supervised Machine Learning algorithms are further divided into single label and multi-label. Single label algorithms are used to assign a document to a single category and multi-label algorithms work on assigning a document to more than one category [8].

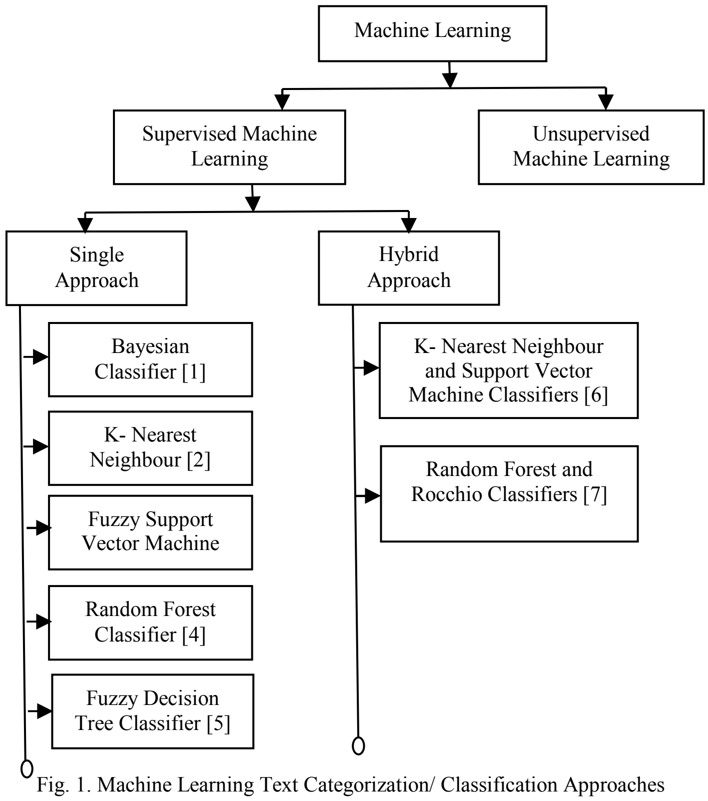

Fig. 1. Machine Learning Text Categorization/ Classification Approaches

On the other hand, Unsupervised Machine Learning aims to cluster the documents without extra knowledge having no human assistance or intervention in such a way that the documents inside a cluster are more alike than documents amid clusters [12]. In this paper, we present Supervised Machine Learning algorithms to classify text. We have considered importance of Feature Selection in Text

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Categorization. These features classify text easily as redundant entities are removed and important features of data are selected and preserved which are then supplied for categorization [1, 3]. Feature Selection help to reduce dimensionality of data. Supervised Machine Learning Text Categorization Approaches are generally divided into two types namely Single approach and Hybrid approach. Single approaches are those which classify text in general having one type of algorithm. One of the approaches for Text Categorization is Bayesian classifier that uses selective features for classification [1]. Another approach is to classify objects into predefined classes of a sample group and select the most similar objects from it known as K- Nearest Neighbour (KNN) [2]. The training data are categorized into subsets that have same class labels containing nodes [3]. Random Forest algorithm deals with classifying based on results determined by random ideas [4]. The other approach can be Integrated Learning model based on Fuzzy Decision Tree [5]. The combination of two or more approaches leads to better performance and speed of Text Categorization. Hybrid approaches covers the benefit of two or more approaches used in it. The high noise tolerance approach also called hybrid approach is also presented in which text encoding is done called two phase SVM and KNN classifier [6]. Another hybrid approach involves the combination of Random Forest and Rocchio algorithm containing benefits of both the classifiers for effective classification. Fig-1 depicts the types of Machine Learning Algorithms that are further divided into Single and Hybrid approaches. Single and Hybrid approaches are further divided into existing categorization or classification algorithms. This paper aims to make review of Supervised Machine Learning approaches for Text Categorization or Classification. Furthermore, we highlight parametric comparison of Text Categorization approaches.

The rest of the paper is structured as follows: The second section represents existing methods, third section represents analysis of methods with their feature based parametric comparison in tabular format and fourth section gives conclusion of the paper.

II. EXISTING METHODS

There are basically two kinds of Text Categorization approaches. They are classified into Single and Hybrid approach. Both the types of approaches are discussed in this section:

A. Feature Selection Based Bayesian Classification

This approach is used for automatic text categorization. It selects the precise feature subset for each class. These feature subsets are called class specific features. Naïve Bayes or Bayesian classification classifies documents having naïve assumption of certain features. This approach contains the combination of Bayes classifier and class specific feature selection. There are basically two types of classes namely reference class which consist of all the documents and a class ci that consist of different features where i=1,2,3 and so on. For each class, important parameters are estimated for classification. Classification rule is built through Baggentoss's PDF Projection Theorem (PPT). Baggentoss's PPT enables projecting probability density functions from a lower dimension feature space to raw data space [13,14]. Conversion into raw data space is required for fair comparison of class specific features. Feature selection like information gain (IG), Chi-Square is done for training dataset for each class which in turn increases performance. IG allows making classification decision on the class to measure information that it contributes in the term's absence or presence. Chi-Square examines the independence of two similar or different events. Fig. 2. shows steps of this approach. This approach is proved to be effective in wide applications of Text Categorization and uses two datasets namely newsgroups [22] and reuters-10 [22]. Feature selection criteria of IG and MD are used. Performance is measured in terms of F-measure and G-mean. These measures are the combination of recall and precision. The author have experimented and obtained overall accuracy of 80% [1].

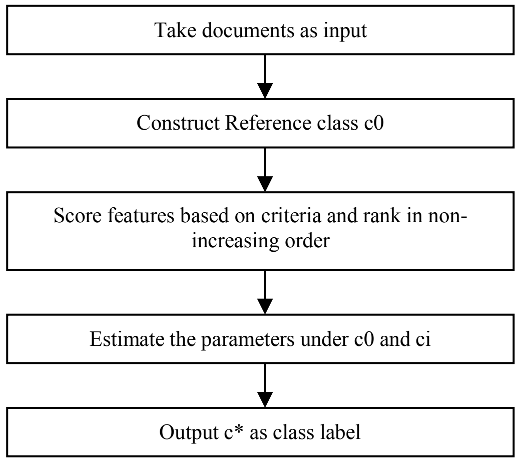

Fig. 2. Steps for Bayesian Classification using class specific features

B. K Nearest Neighbour

K Nearest Neighbour is a lazy learning popular algorithm. This approach combines KNN with TF-IDF (Term Frequency Inverse- Document Frequency) method for Text Classification [2].

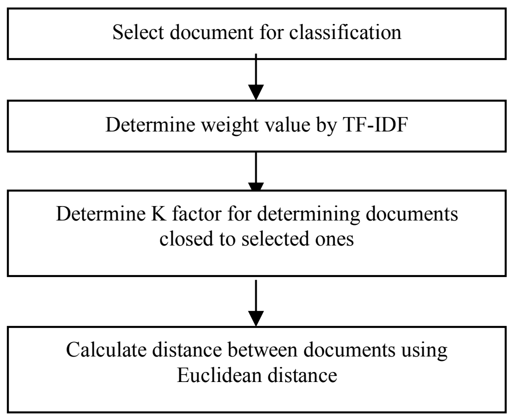

Fig. 3. Steps for KNN classifier

TF-IDF is a numerical statistical method that allows determining weight for each term in each document. KNN doesn't require training data to perform classification [15]. It finds the most similar objects from sample groups by mutual Euclidean distance. K value of KNN determines required number of documents from the collection which is closest to the selected document. This makes the classifier more reliable and good results are achieved irrespective of K factor value. The performance is dependent on quality of documents and words inside it. If they are unused usually then the performance is degraded which is why some preprocessing is required before doing the classification. Accuracy adheres from 65% to 90%. The data is taken online having 500 documents of different categories. Fig. 3. depicts the steps for KNN classifier.

C. Fuzzy Support Vector Machine

This approach came into existence to bury the limitation of one vs. all Support Vector Machine (SVM). The problem with this method was that data may be classified into an undefined class. Fuzzy SVM approach defines the associated membership function for each multi-labeled class and a data point is classified into a class which possesses the largest membership function [16,17]. Moreover, accuracy can be improved as compared to one vs. all approach. This approach uses heuristic to solve unclassified regions. Benchmark datasets of 12 different types were used to experiment the accuracy of this technique by the author. A high accuracy is obtained but differs for various types of datasets. This technique also solves unclassified regions [3].

D. Random Forest

This is an integrated learning algorithm. Integrated Learning determines category of target samples with improved accuracy and generalization by combining the basic multiple categorizers and analyzing results [18].

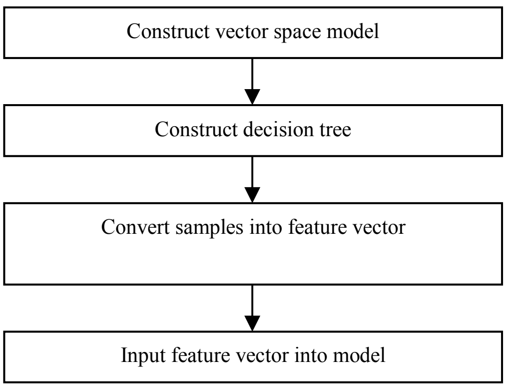

Fig. 4. Steps for Random Forest

It uses K decision tree as a basic categorizer. The result is determined by random ideas like bagging and feature subspace. Training set and attribute subset are selected randomly in constructing a decision tree. It can be said that Random Forest has generalization ability because generalization error develops towards upper bound. First the vector space model is constructed using word segmentation. Decision tree is then constructed using Bagging and CART algorithms. Bagging is used to select training samples from original sample. CART builds the decision tree optimally. The testing samples are converted into feature vector and then inputted into the model. The output from this step gives the classification of samples. The steps of this classifier is shown in Fig. 4. This approach gives good performance in Text Classification. Sogou laboratory text categorization corpus is used as dataset which contain five different categories. Feature selection is done in this technique using information gain. Performance is calculated using the combination formula of precision and recall termed as F1 measure. After experimenting by the author, this technique possess 89-90% accuracy [4].

E. Fuzzy Decision Tree

This is a well-known top down approach which has certain features like higher comparative accuracy, doesn't require large number of parameters to be adjusted and easy due to intuitive topology. Decision trees are divided into classic, fuzzy and oblique. The training data are partitioned into subsets having similar class labels and terminated when all data associated with a node belong to the same class [19, 20]. In this, the addition of an impure node is added for those samples which are not assigned to the classes with fuzzy rule. New fuzzy rules are applied on those impure node which gives rise to new leaf node. This continues till the tree grows and fuzzy rules control the expansion of tree. Dataset used is UCI repository [22] and synthetic datasets. Due to feature selection, the fuzzy rule based decision tree performs accurate in terms of scores of effectiveness as compared to conventional decision tree when experimented by author [5].

F. K Nearest Neighbour and Support Vector Machine

This approach integrates Nearest Neighbour Classification and Support Vector Machine approach. The goal is to reduce the impact of parameters in classification accuracy [21].

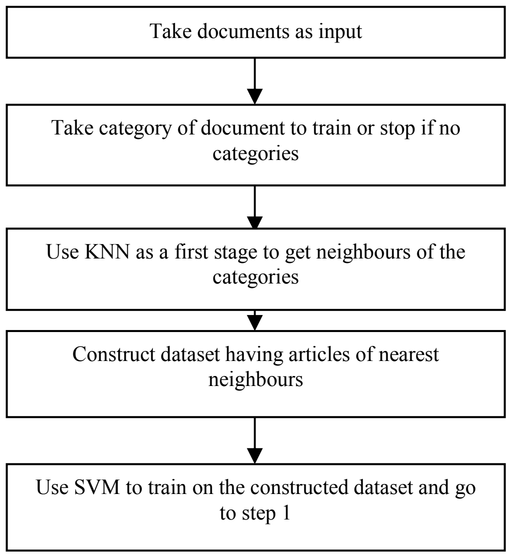

Fig. 5. Steps for hybrid KNN and SVM algorithm

Initially SVM reduces training samples of the available categories to their support vectors. These support vectors are used as training data in Nearest Neighbour algorithm. From this, similarity measure and distance function is used to calculate belonging of testing data to a particular class. This eventually reduces time consumption. The initial use of KNN and later use of SVM proves to be scalable. SVM and KNN when combined gives good results which is experimented by the author and derived that the performance score of 42.72% is obtained. Moreover, this performance score is very good as large scale dataset is being used [6].

G. Random Forest and Rocchio Algorithm

This approach combines Rocchio and Random Forest algorithm and performs multi-label text categorization. Pre-processing techniques like stop word remover and word stemmer are incorporated to overcome disadvantage of Rocchio algorithm. To reduce the error rate, Random forest algorithm takes minimal categories as input. Dimensionality is reduced in this approach.

Key value pair of word and term frequency is generated respectively. Cosine similarity value is found using vector space model which is taken as threshold value. Categories having cosine similarity value are chosen for input to Random forest algorithm which chooses a random subset of categories and create a set of decision trees. After validating the testing documents, positive results which are greater than negative results are stored as relevant categories. These are updated using Rocchio algorithm. In this approach, training data set vector is generated only once and training time is reduced. The methodology is depicted in Fig. 6. Results show that accuracy is high. Accuracy is used to measure the performance of this hybrid technique. It is given as the ratio of true positive over sum of true positive and false positive results of samples. Datasets used are 20 Newsgroup [22] and RCV1 [22]. The experiments done by author show that accuracy is obtained from 85 to 98% depending upon the categories [7].

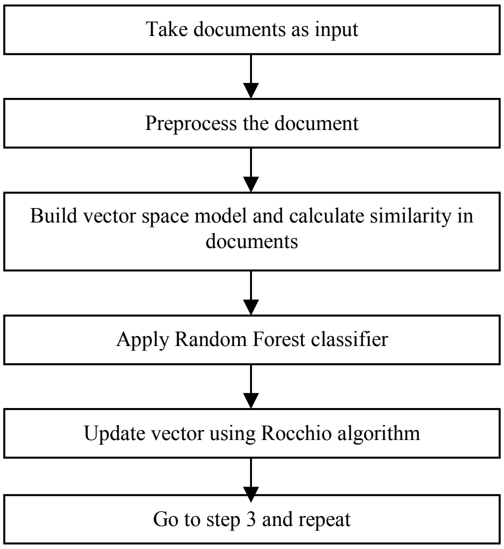

Fig. 6. Hybrid Approach involving Random Forest and Rocchio Algorithm

III. PARAMETRIC COMPARISON

There are different Machine Learning algorithms which can be classified into supervised and unsupervised. In this section we analyze different parameters of Supervised Machine Learning Text Categorization approaches to classify text. Supervised Machine Learning contains some predefined dataset or prior information which can be used to classify text in the later stage. Various parameters and their significance are discussed as under:

- Approach type - This signifies the type of text categorization approach. It can be classified into single and hybrid. Single text categorization approach uses single type of algorithm to classify text while hybrid text categorization approach uses multiple algorithms and strategies or there is a mixture of two or more text categorization approach to classify the text or documents. It is observed that hybrid text classifiers are more flexible and performance proof.

- Single/ Multi-label - Single label indicates that the instance of a text can be assigned to classes based on some constraints and not beyond than that, while in multi-label type of classification, the instance of the text can be assigned to more than one classes without any constraints.

- Feature Selection Consideration - Due to increase in dimensionality of text, feature selection is done to pick out relevant features and remove unwanted features from text before classification of text [9]. This stage is used by many supervised machine learning algorithms while classifying text. This parameter if used then it reduces the complexity and overhead of the algorithms to classify text.

- Learning model - This can be classified into eager and lazy learning methods. In Eager learning method system constructs a general target function which is not dependent on input during the training of the system. Its main advantage is that the target function can be approximated globally during the training of the system and requires much less space. It also deals better with noise. However, in Lazy learning method generalization beyond training data is delayed. In this, the target function is approximated locally thus it can simultaneously solve multiple problems and deal with the changes in problem domain easily.

- Use of Strategy - All the available algorithms uses different strategies to build a pattern through which Text Categorization can be done easily and quickly. Basic assumption which algorithms take while processing any classification techniques is very important to measure any metrics like performance, cost and so on.

- Linear/ Non-linear - An approach is said to be a linear approach if it doesn't require building any hierarchy to classify text. A linear approach can be considered as a simple approach where data is classified in a linear manner, while in non-linear approach, text is classified by building a hierarchical tree like structure which can be divided into various nodes making a tree and resulting into the classification of text. The linear or non-linear type of approach also affects the performance and time of an algorithm to classify text.

easy. This allows reduction in time to classify the text [8].

- Over-fitting - Over-fitting means a model that models the data too well. It learns the text data into so much of extent that it degrades the performance and results in negative impact. In general, the text data must be between the over and under-fitting of the approaches.

TABLE-1 PARAMETRIC COMPARISON OF SUPERVISED MACHINE LEARNING TEXT CATEGORIZATION APPROACH

Method / Parameter	Class specific Bayesian	KNN	Fuzzy SVM	Random Forest	Decision tree	KNN and SVM	Random forest and Rocchio
Single/ hybrid	Single	Single	Single	Single	Single	Hybrid	Hybrid
Applied to single label/multi-label	Single label	Single label	Multi-label	Single label	Multi-label	Single label	Multi-label
Feature selection done	Yes	No	Yes	No	Yes	No	No
Learning model	Eager	Lazy	Eager	Integrated	Eager	Eager	-
Based on	Naïve assumption of features	Similar objects from sample groups	Kernal	K decision tree	Fuzzy rules	One vs near [two stage approach]	Cosine similarity and decision tree
Linear/nonlinear/ both	Linear	Non-linear	Linear	Non-linear	Non-linear	Both	Both
Noise tolerancy	High	Low	Moderate	Moderate	High	Moderate	Moderate
Provide transparency	No	Yes	No	No	Yes	May be	No
Ovefitting problem [yes/no]	No	Yes	No	No	Yes	No	No

- Noise Tolerancy - This is a very crucial parameter for text classification. Noise can be defined as any outside entity that combines with the text and disturbs the content resulting into unrecognizable original text. Many supervised machine learning approaches are prone to noise which leads to data loss or unidentified data as the format or the content gets disrupted. Many approaches exist that protects the data from noise.

- Transparency - This parameter indicate what level of transparency the Text Categorization approach is providing for classifying text. Some approaches provide full transparency while other doesn't. Classification time is strongly dependent on this parameter. If the text displayed is up to the mark and consistent and precise then it takes less time to classify the text otherwise the time duration may increase. It means that the texts are displayed in a high level of transparency where understanding becomes

A fit can be defined as how well we approximate the target function. Under-fitting machine learning model is not able to model or generalize the data further. Thus the data should be a good fit. The performance can degrade if the data is under-fitted or over-fitted.

We analyze the Supervised Machine Learning Text Categorization approaches in this section in Table-1 where we compare the approaches with respect to different parameters that we discussed above in detail. This analysis is done on the basis of certain good and bad features of Text Categorization approaches. Table-1 showcases the parametric evaluation of seven existing Supervised Machine learning text categorization approaches. The table describes different parameters that can be used to evaluate Text Categorization approaches for classification of text. These approaches are considered in such a way that we can derive an idea how efficient they are to classify text.

IV. CONCLUSION

Text can be classified into different sub texts using Supervised Machine Learning approaches. Different approaches exist that classify text so that the quality is maintained and time duration to classify the text is lessened. We have also seen that single and hybrid approaches exist to classify text. We observe that approaches which are hybrid, good fit and provide proper level of transparency can be considered best to classify text. We also observed that the approach which uses feature selection as the initial stage gives good result. In this paper, the hybrid algorithms called Random and Rocchio algorithm fulfils the above criteria and such two stage approach can be used to classify text in a better way. It is also observed that the limitations of Rocchio algorithm can be removed and along with Rocchio, the text classification becomes easy. The only problem with the above approach is that the time limit is not maintained. This issue if overcomes then it becomes one of the best approaches to classify text.

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